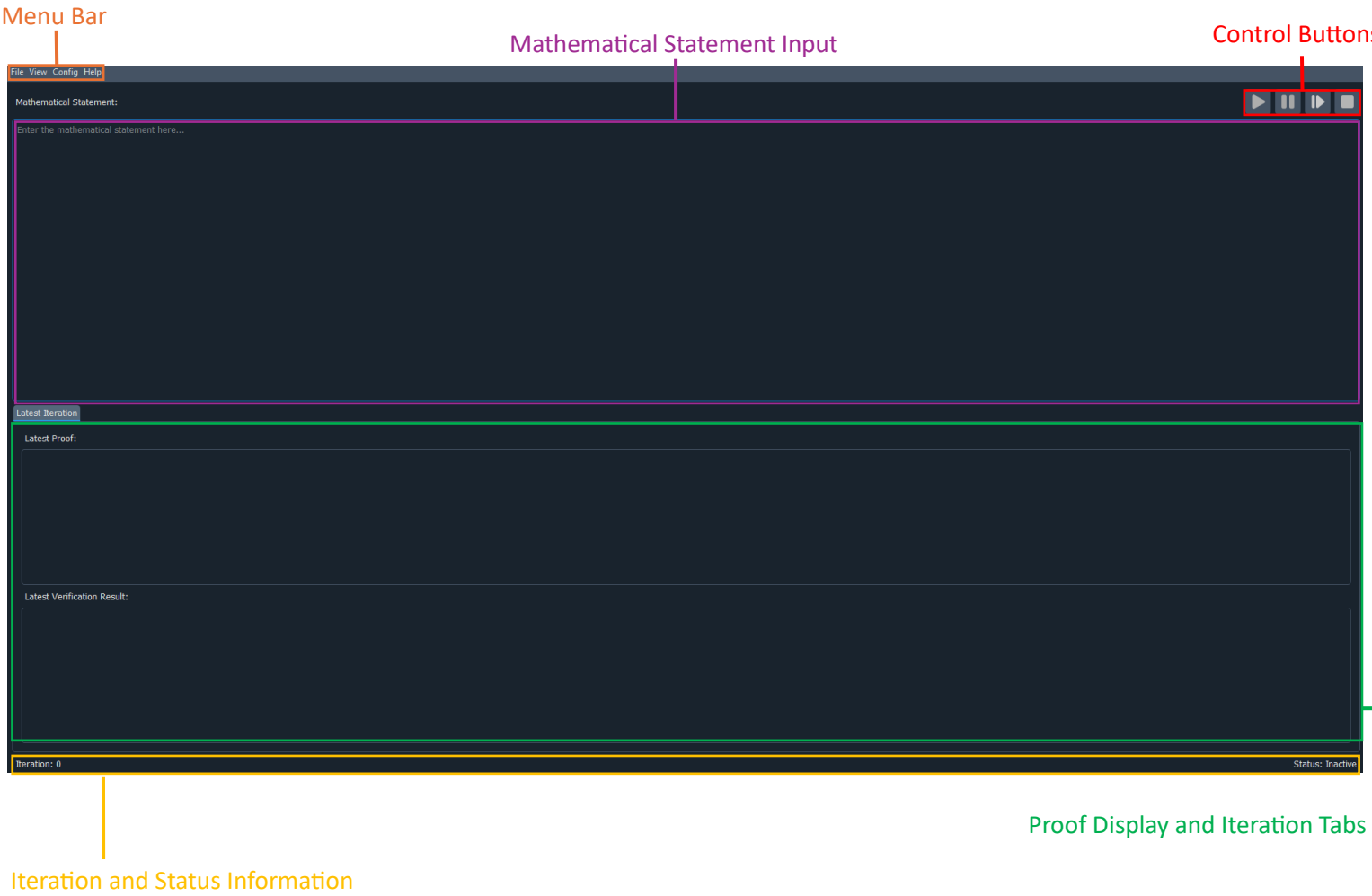


Proof Verifier – User Help

Welcome to Proof Verifier - a GUI-based tool designed for the interactive generation and verification of mathematical proofs. It allows users to input mathematical statements, initiate proof verification, and monitor iterative refinement processes with detailed outputs displayed in multiple tabs.

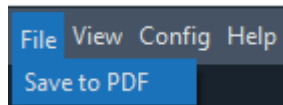
GUI Overview:



Menu Bar:

1. File

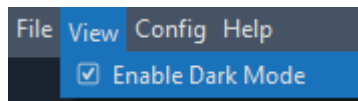
- Save to PDF



Downloads automatically to downloads folder a summary of the entire run, highlighting the number of iterations, the result, the original statement and each iteration separately.

2. View

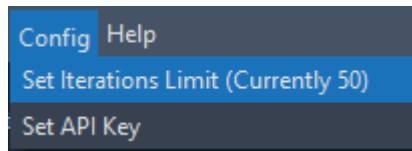
- Enable Dark Mode



Toggles the appearance, such as enabling or disabling dark mode (defaulted to dark mode, chosen mode will be saved in configuration for future uses).

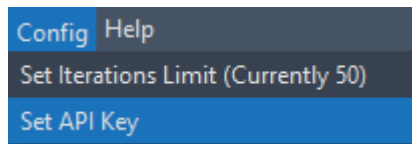
3. Config

- Set Iteration Limit



Sets iteration limit for proof verification (Will be saved under <USER>\ProofVerifier for any operation system).

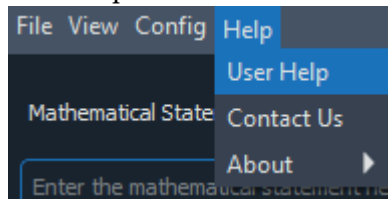
- Set Iteration Limit



Sets the API Key. This step is mandatory and the verifier will not start before setting a validate OpenAI API Key (Will be saved under <USER>\ProofVerifier for any operation system).

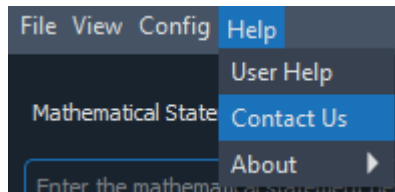
4. Help

- User Help



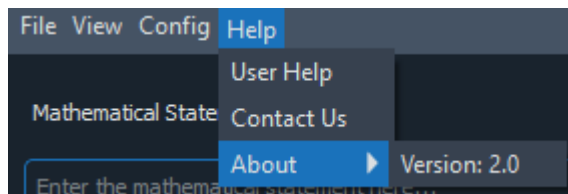
Shows the current PDF document, offers guidance on using the application.

- Contact Us



Shows the current PDF document, offers guidance on using the application.

- About



Displays current system version and credits.

Mathematical Statement Input

In this section, the user can input a mathematical proof, and our system will try to validate its correctness by iteratively refining the proof using ChatGPT's API.

Mathematical Statement:

Euclid's Theorem

▶

⏸

▶

□

Once the user will input a statement for verification, the “Verify” button will be enabled.
After hitting verify, the iterative proof verification process will begin.

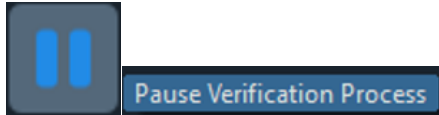
Control Buttons

Proof Verifier has 4 control buttons for best user experience.

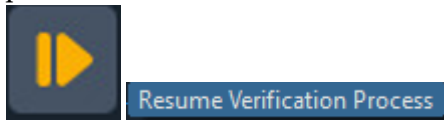
1. Verify Button – Initiate the Verification (enabled when verification process is inactive and when there's a statement to verify).



2. Pause Button – Pause the Verification (Will be paused at the end of the current iteration, enabled when the verification process is active).



3. Resume Button – Resume the Verification (enabled only when the verification process is paused).



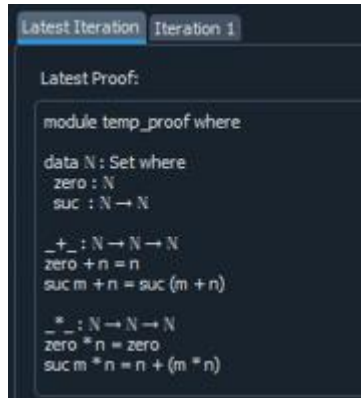
4. Stop Button – Stop the iteration (Will be stopped at the end of the current iteration, enabled when the verification is paused or active).



Proof Display and Iteration Tabs

To view the progress of the verification process, proof verification provides a Proof Section (Agda code), Verification Result (feedback from Agda compiler), tabs for each iteration.

1. Proof Section – In this section, the user can see the proof generated by ChatGPT API in the Latest iteration.



```
Latest Iteration  Iteration 1

Latest Proof:

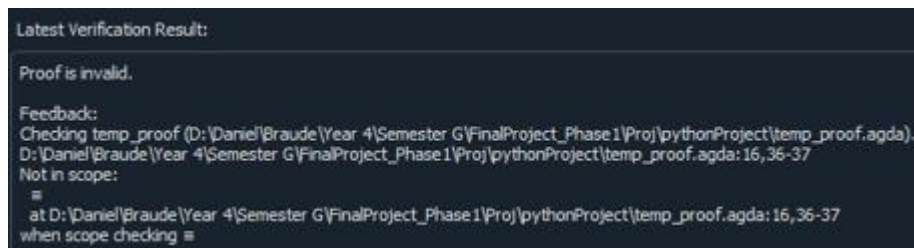
module temp_proof where

data N : Set where
  zero : N
  suc  : N → N

_+_ : N → N → N
zero + n = n
suc m + n = suc (m + n)

_*_ : N → N → N
zero * n = zero
suc m * n = n + (m * n)
```

2. Verification Result - In this section, the user can see the feedback from Agda compiler in the Latest iteration.

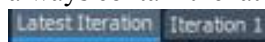


```
Latest Verification Result:

Proof is invalid.

Feedback:
Checking temp_proof (D:\Daniel\Braude\Year 4\Semester G\FinalProject_Phase1\Proj\pythonProject\temp_proof.agda).
D:\Daniel\Braude\Year 4\Semester G\FinalProject_Phase1\Proj\pythonProject\temp_proof.agda:16,36-37
Not in scope:
=
at D:\Daniel\Braude\Year 4\Semester G\FinalProject_Phase1\Proj\pythonProject\temp_proof.agda:16,36-37
when scope checking =
```

3. Tabs – Each refinement will get its own iteration tab with the addition of a latest tab which will always contain the latest refinement.

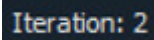


```
Latest Iteration  Iteration 1
```

Iteration and Status Information

At the bottom of the screen, there is information about the current iteration and the status of the tool.

1. Iteration Info – displayed at the bottom left side of the screen.



Iteration: 2

2. Tool Status – displayed at the bottom right side of the screen, the following statuses exist:
 - a. “Inactive” – When the verification process is finished or hasn’t started.
 - b. “Active” – When the verification process is active.
 - c. “Pausing...” – When the verification process is being paused.
 - d. “Paused” – When the verification process has been paused.
 - e. “Stopping...” – When the verification process is being stopped.
 - f. “Stopped” – When the verification process has been stopped.

Example Workflow

1. Enter a mathematical statement in the input box.
2. Click the Verify button to initiate proof verification.
3. Monitor the iterative process through the tabs and status label.
4. Pause, resume, or stop the process as needed.
5. View results in the Latest Iteration tab or review detailed feedback in previous iteration tabs or alternatively download a pdf file with a detailed summary of the run.

Cases where the verifier ceases its job

1. A valid proof has been found – A success message will be displayed.
2. Iterations limit has been reached – A failure message will be displayed.